1. *What is Parkinson's disease?*

★ My Understanding:

- Parkinson's disease is a progressive neurological disease that involves the death of dopamine-producing nerve cells. Dopamine is a powerful neurochemical that plays a major role in movement, and the death of the nerve cells that produces them results in flaws in the movement of a patient, ranging from tremors to slow reaction.

★ Definition:

- Parkinson's disease is a progressive, incurable movement disorder of the nervous system. Symptoms begin as barely noticeable tremors and slight stiffness, evolving into slurred speech and slowed movement (*bradykinesia*).

2. *What are the numbers?*

★ According to the Government of Canada, sourcing data from the 2022-2023 fiscal year, 530 of 100,000 Canadians over the age of 40 lived with some form of Parkinsonism, a category of movement disorders, of which 3 out of 5 were men. Around 38 Canadians are diagnosed with Parkinsonism daily, with 8 out of 10 Canadians over the age of 65 being diagnosed.

★ In my home province of Nova Scotia, between 2015 and 2025, nearly a decade, cases of diagnosed Parkinsonism remained steady between 2015 and 2024, in both sexes over the age of 40. However, it is important to keep in mind that wait times for a diagnosis can account for multiple unrecorded numbers.

3. *What are the symptoms?*

★ Bradykinesia: In simple terms, this is slowed movement. Bradykinesia ranges from struggling to show facial expressions to having difficulty blinking.

★ Freezing of Gait: This is a sudden, involuntary inability to move the feet in the event of activities such as walking down a narrow hallway. It rarely occurs during clinical evaluations, and it normally lasts a few seconds.

★ Anosmia: An olfactory dysfunction, anosmia typically occurs 10 to 20 years prior to a formal PD diagnosis.

★ REM Sleep Behavior Disorder: During the *Rapid Eye Movement* stage of sleep, the brain temporarily paralyzes the body, a state known as muscle atonia. This prevents the body from physically acting out vivid dreams. REM Sleep Behavior Disorder is chronologically similar to anosmia in that it occurs a decade or two prior to a formal PD diagnosis. The loss of muscle atonia leads to patients engaging in a range of actions during dreams, from simple twitches to more complex actions, such as kicking and fighting.

## *4. Why Dopamine?*

Dopamine is a monoamine neurotransmitter and neurohormone. As a neurotransmitter, it acts as a "reward center" for many bodily functions. These functions are based on movement, memory, and motivation. As a neurohormone, dopamine is peripherally responsible for the vasodilation and vasoconstriction of blood vessels, regulating the removal of sodium and urine from the body, reducing the amount of insulin produced by the pancreas, slowing the movement of contents in the gastrointestinal (GI) tract, and reducing lymphocyte activity in the immune system.

As it is responsible for movement and motivation, the depletion of dopamine results in poor movement (tremors, bradykinesia) and an increase in apathy, which is why many PD patients can often "do nothing" for some time without feeling positive or negative about the inactivity. The depletion of dopamine is the cause of many neurological disorders, which is why PD will often coincide with other conditions, such as dementia. Furthermore, the depletion of dopamine leads to a loss in the quality of its functions, such as blood vessel and insulin regulation, which become immediate effects for patients.